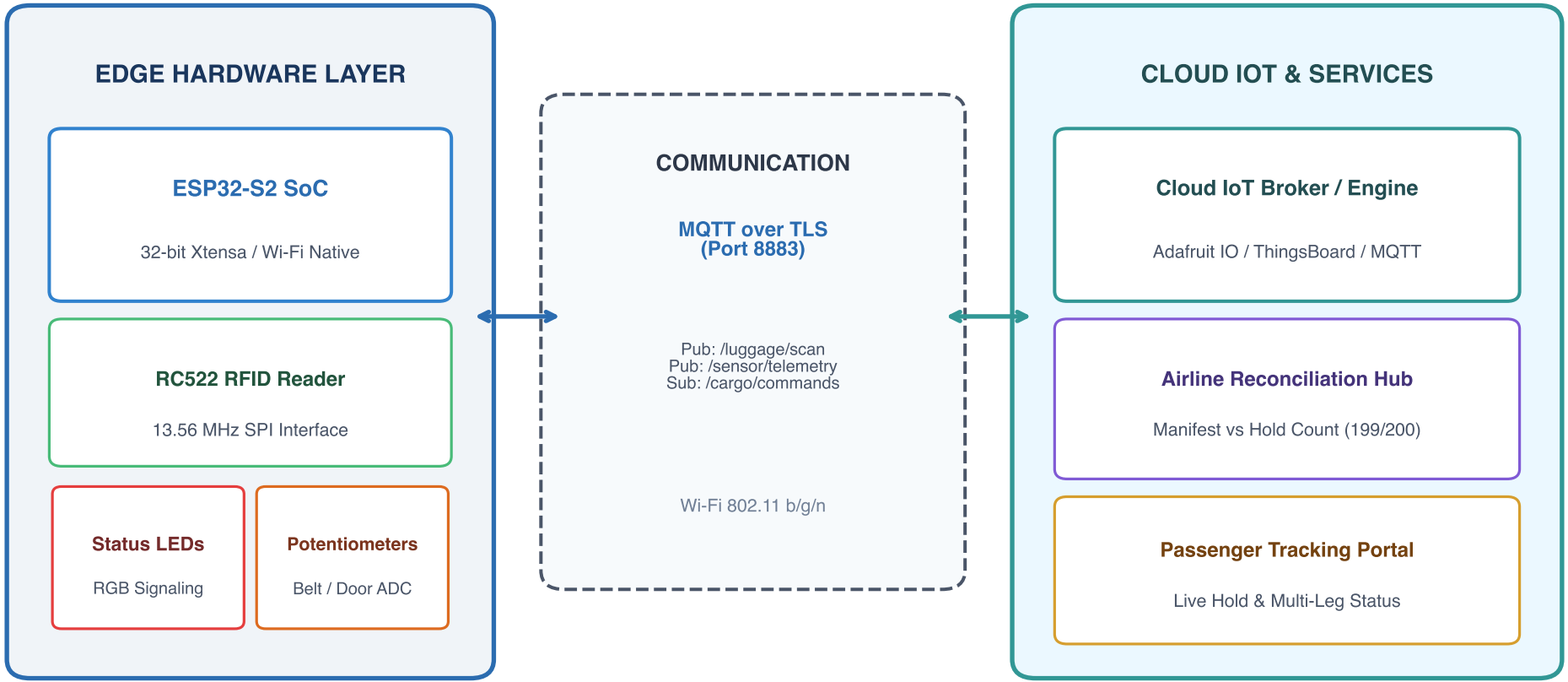
EDGE HARDWARE LAYER
ESP32-S2 SoC
32-bit Xtensa / Wi-Fi Native
RC522 RFID Reader
13.56 MHz SPI Interface
Status LEDs
RGB Signaling
Potentiometers
Belt / Door ADC
COMMUNICATION
MQTT over TLS
(Port 8883)
Pub: /luggage/scan
Pub: /sensor/telemetry
Sub: /cargo/commands
Wi-Fi 802.11 b/g/n
CLOUD IOT & SERVICES
Cloud IoT Broker / Engine
Adafruit IO / ThingsBoard / MQTT
Airline Reconciliation Hub
Manifest vs Hold Count (199/200)
Passenger Tracking Portal
Live Hold & Multi-Leg Status